

Oluwatobi Melvyn Mayungbo

Data Scientist, Statistics and Forecasting

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SUMMARY

Statistics graduate who builds and tests data-driven models in Python. Built sequence models over transaction histories judged across 3 seeds, a crypto risk scorer across five chains, a causal study with bootstrap intervals, and rolling-origin backtests with leakage tests that fail the build.

SKILLS

Statistics and Causal Inference: Statistical Inference, Time-Series Backtesting, Experiment Design, Causal Inference (S/T/X-Learners, Causal Forests), Power Analysis, CUPED, Uplift Modeling, Qini and AUUC, Logistic Regression, WOE Scorecards

Machine Learning: PyTorch, LightGBM, Sequence Models (GRU, TCN, Transformers), Scikit-learn, XGBoost, Deep Learning, Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Payments and Risk: Web3 Fraud Risk Scoring, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting

Engineering and MLOps: Python, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.
- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type

breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.

Uplift Modelling for Campaign Targeting | [GitHub](#)

Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions

- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action